Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

7 November 2024
8:45-11:45h

This exam consists of 9 exercises for 202001172 and of 11 exercises for 202001171.
Motivate all your answers.

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1. Find
$$\lim_{x\to\infty}\left(\sqrt{x^2+x}-\sqrt{x^2-x}\right)$$

2. Find the derivative of
$$y=\frac{x^2+3e^x}{2e^x-x}$$

3. Find the extreme values (absolute and local) of the function $y=x^2\ln x$. Also write down where the extreme values occur.

4. Evaluate the following integrals

   (a)
   $$\int \frac{1}{\sqrt{t}}\cos(\sqrt{t}+3)\,dt$$

   (b)
   $$\int \frac{x+3}{2x^3-8x}\,dx$$

5. Find the length and direction of $\mathbf{u}\times\mathbf{v}$ where
$$\mathbf{u}=\frac{3}{2}\mathbf{i}-\frac{1}{2}\mathbf{j}+\mathbf{k} \text{ and } \mathbf{v}=\mathbf{i}+\mathbf{j}+2\mathbf{k}$$

6. Replace the following polar equation with its equivalent Cartesian equation. Then describe the graph.
$$r^2=4r\sin\theta$$

7. Find parametric equations for the line that is tangent to the following curve at the given parameter value $t=t_0$.
$$\mathbf{r}(t)=(\cos t)\mathbf{i}+(\sin t)\mathbf{j}+(\sin 2t)\mathbf{k},\ t_0=\frac{\pi}{2}$$

8. Express $\partial z/\partial u$ in terms of u and v if

   $z=\sin xy+x\sin y$
   $x=u^2+v^2$
   $y=uv$

   Then evaluate $\partial z/\partial u$ at $u=0, v=1$.

PLEASE TURN OVER

9. Solve the following initial value problem

$$t\frac{dy}{dt} + 2y = t^3, \;\; t > 0, \;\; y(2) = 1$$

Exercises 10 and 11 are only for 202001171.

10. Find all solutions of the equation $x^4 + 4x^2 + 16 = 0$.

11. Find the unique solution of the following second-order initial value problem.

$$y'' - 4y' + 4y = 0, \;\; y(0) = 1, \;\; y'(0) = 0$$

Norm:

Exercise	1	2	3	4a	4b	5	6	7	8	9	10	11	total
202001171	4	4	3	3	4	3	3	4	4	4	5	4	45
202001172	4	4	3	3	4	3	3	4	4	4	-	-	36

Exam grade 202001171: 1 + (obtained points)/5
Exam grade 202001172: 1 + (obtained points)/4